

Lab Work: JSON Schema and Validation

BE in Information Technology · Unit I: Data Mapping and Exchange · 25 marks

Scenario

A library keeps its holdings in a digital catalogue. Each entry in the catalogue describes a single book as one JSON object. Your task is to define what a valid catalogue entry looks like, produce data that does and does not conform to it, and check that data automatically.

Task 1 — Design the schema

8 marks

Write a JSON Schema, saved as `catalogue.schema.json`, that describes a single library catalogue entry according to the following rules:

- `isbn` — a string of exactly 13 digits.
- `title` — a non-empty string.
- `authors` — an array of strings containing at least one author.
- `year` — a whole number between 1450 and 2025 inclusive.
- `format` — one of: `hardcover`, `paperback`, `ebook`, `audiobook`.
- `available` — a boolean.
- `isbn`, `title`, `authors`, `year` and `format` must always be present; `available` is optional.
- No fields other than the seven listed above are permitted in an entry.

Task 2 — A conforming entry

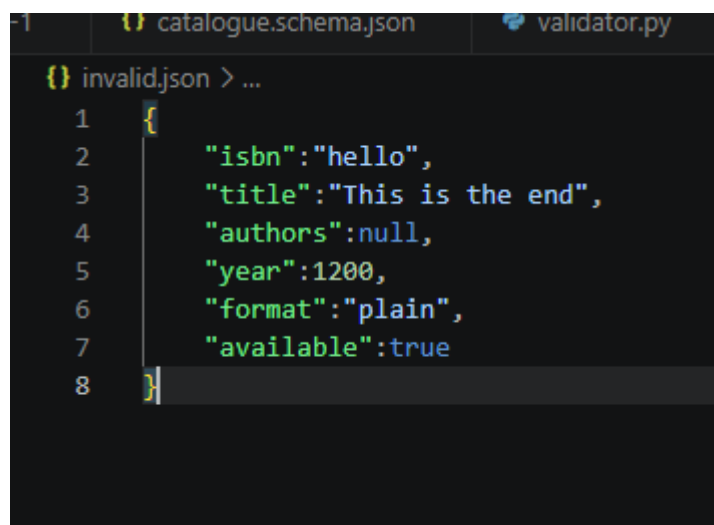
3 marks

Create a file `valid.json` holding one catalogue entry that satisfies every rule in your schema.

Task 3 — A non-conforming entry

5 marks

Create a file `invalid.json` holding one catalogue entry that breaks at least four different rules from Task 1. Alongside the file, state which rule each error violates.



```
1 {} invalid.json > ...
2 {
3   "isbn":"hello",
4   "title":"This is the end",
5   "authors":null,
6   "year":1200,
7   "format":"plain",
8   "available":true
}
```

1. `isbn` isn't exact 13 digits.
2. There isn't any author, it is required, not optional.
3. Year is not between the required boundary

4. The format is not the one specified in the list of format in the json schema.

Task 4 — A validator program

7 marks

Write a program in either JavaScript or Python that loads `catalogue.schema.json` and validates both `valid.json` and `invalid.json` against it. Use a schema-validation library — Ajv for JavaScript or the package `jsonschema` for Python. For each file, the program must report whether it passed and, where it failed, the validation errors returned by the library.

Task 5 — Evidence of the run

2 marks

Capture the output of your program showing `valid.json` passing and `invalid.json` failing with its reported errors.

```
validator.py > ...
1  import json
2  from jsonschema import validate, ValidationError
3
4  with open("catalogue.schema.json", "r") as schema_file:
5      schema = json.load(schema_file)
6
7  with open("valid.json", "r") as data_file:
8      data = json.load(data_file)
9
10 try:
11     validate(instance=data, schema=schema)
12     print("Success: Data matches the schema!")
13 except ValidationError as e:
14     print(f"Validation Error: {e.message}")
15
```

```
PS C:\Users\DELL> Open file in editor (ctrl + click) \Users\DELL\AppData\Local\Programs\Python\Python314\python.exe c:/Users/DELL/Desktop/02230147/validator.py
Success: Data matches the schema!
PS C:\Users\DELL\Desktop\02230147>
```

```
validator.py > ...
1  import json
2  from jsonschema import validate, ValidationError
3
4  with open("catalogue.schema.json", "r") as schema_file:
5      schema = json.load(schema_file)
6
7  with open("invalid.json", "r") as data_file:
8      data = json.load(data_file)
9
10 try:
11     validate(instance=data, schema=schema)
12     print("Success: Data matches the schema!")
13 except ValidationError as e:
14     print(f"Validation Error: {e.message}")
15
```

```
PS C:\Users\DELL\Desktop\02230147> & C:\Users\DELL\AppData\Local\Programs\
\python.exe c:/Users/DELL/Desktop/02230147/validator.py
Validation Error: 1200 is less than or equal to the minimum of 1450
PS C:\Users\DELL\Desktop\02230147>
```

Submit

catalogue.schema.json, valid.json, invalid.json, your validator source file, and the captured output — combined into a single report.

Marks

Task	Marks
Task 1 — Design the schema	8
Task 2 — A conforming entry	3
Task 3 — A non-conforming entry	5
Task 4 — A validator program	7
Task 5 — Evidence of the run	2
Total	25